

HealthConnect Clinic – Week 6 Project Summary

Track: Data Analytics

Project: HealthConnect Clinic

1. Week 6 Objective

The objective of Week 6 was to build on the HealthConnect Week 5 Data Analytics work by moving from initial analysis and dashboard development into advanced analysis, validation, integration and decision support.

The focus was to validate important Week 5 findings, investigate deeper relationships and segments, improve the existing dashboard, support business recommendations and contribute analytical findings to the Data Science track.

2. Planned Week 6 Activities

The planned Data Analytics activities were to:

- investigate the most important Week 5 findings;
- validate whether the findings remained meaningful;
- analyse important patient and appointment segments;
- investigate relationships associated with no-show behaviour;
- validate Week 5 KPIs;
- identify findings with the greatest business impact;
- translate findings into practical HealthConnect actions;
- provide relevant findings to Data Science;
- review and refine Week 5 conclusions;
- improve the Week 5 dashboard;
- document evidence supporting recommendations; and
- document the cross-track analytical contribution.

3. Work Completed

The Week 6 Data Analytics work was completed across the planned analytical activities.

The three main findings selected for deeper investigation were:

1. Booking lead time and no-show behaviour.
2. Previous no-show history and future no-show behaviour.
3. Reminder status and no-show behaviour.

Additional segment and relationship analysis was then conducted to determine how these findings behaved across appointment types and when multiple variables were considered together.

The Week 5 KPIs were also revalidated and remained accurate.

4. Improvements From Week 5

Week 6 improved the Week 5 work by moving beyond the initial descriptive findings.

The major improvements included:

- deeper validation of the main no-show findings;
- additional segment-level analysis;
- combined relationship analysis;
- KPI validation;
- clearer identification of business impact;
- more specific evidence-based recommendations;
- refinement of the interpretation of appointment type;
- improvement of the existing dashboard rather than rebuilding it;
- and cross-track integration with Data Science.
- The Week 5 dashboard was duplicated before Week 6 development so that the original Week 5 work was preserved.

5. Key Analytical Findings

The strongest Week 6 findings were:

- No-show rates increased substantially as booking lead time increased, from 29.01% for 0–9 days to 67.92% for 50–59 days.

- No-show rates increased with previous no-show history, from 43.51% with no previous no-shows to 67.95% with three previous no-shows.
- Appointments with reminders had a 47.36% no-show rate compared with 51.39% without reminders, producing a 4.03 percentage-point gap.
- The main patterns generally remained visible across appointment types.
- Combined analysis showed that booking lead time and previous no-show history provide important behavioural signals.
- Appointment type should be interpreted alongside other factors rather than treated as an isolated explanation for no-show behaviour.

6. Cross-Track Integration

The Data Analytics track collaborated with the Data Science track.

Data Analytics provided validated findings on booking lead time, previous no-show behaviour, appointment type, reminders, combined relationships and KPIs.

Data Science reviewed these findings against its model and error analysis.

The collaboration provided useful confirmation that:

- booking lead time was the strongest model feature;
- historical no-show behaviour was also an important model feature;
- the reminder finding was consistent with the modelling analysis; and
- Specialist Consultation's higher model error rate should not be confused with having the highest actual no-show rate.
- The collaboration also identified appointment type × booking lead time as a potential interaction to investigate further during Week 7.

7. Important Decisions Made

The main decisions made during Week 6 were:

- retain the validated Week 5 KPIs;
- focus deeper analysis on the strongest behavioural findings;

- avoid treating small groups as stable estimates;
- distinguish association from causation;
- preserve the Week 5 dashboard rather than overwrite it;
- use cross-track evidence to refine interpretation;
- carry the appointment type × booking lead-time interaction into Week 7 testing.

8. Major Challenges and Risks

The main challenges were:

- small sample sizes in some segments;
- potential over-interpretation of very high percentages in small groups;
- distinguishing model error from actual appointment risk;
- avoiding causal claims from observational data;
- and ensuring that cross-track findings were interpreted consistently.

These risks were documented and will continue to be monitored during Week 7.

9. Remaining Issues

The main remaining analytical issue is to further test whether the observed relationships remain reliable under additional validation.

Particular attention should be given to:

- appointment type × booking lead time;
- small patient/appointment segments;
- reminder effectiveness;
- KPI accuracy after further development;
- and model uncertainty identified by Data Science.

10. Contribution to the Overall HealthConnect Project

The Week 6 Data Analytics work strengthened the HealthConnect project by providing validated evidence that can support both operational decision-making and predictive modelling.

The findings support targeted attention for long-lead appointments and patients with previous no-shows, while also providing evidence for reviewing reminder processes.

The cross-track integration further connected the Data Analytics findings with the Data Science modelling work, creating a stronger foundation for the next stage of development.

11. Proposed Week 7 Focus

The Data Analytics focus for Week 7 will be testing, validation and refinement.

The main priorities will be to:

- further validate the key no-show relationships;
- test segment consistency;
- recheck KPI accuracy;
- test the improved dashboard;
- review recommendations against new evidence;
- investigate the appointment type × booking lead-time interaction; and
- continue monitoring small sample sizes and analytical limitations.

12. Overall Week 6 Summary

Week 6 successfully progressed the HealthConnect Data Analytics work from the Week 5 foundation into advanced analysis, validation and decision support.

The major Week 5 findings remained meaningful after deeper investigation, the core KPIs were validated, the dashboard was improved without discarding the original work, and meaningful analytical integration was completed with Data Science.

The project is therefore positioned for Week 7 testing and refinement, rather than requiring the complete HealthConnect solution to be finished at this stage.